

Relation Algebra

§10. Freyd’s Power allegories

(10a)

A **power allegory** is a division allegory with one unary operation on arrows, $\exists$ **epsilon**, subject to

$\exists_R \square = R \square, \quad \exists_R = \exists_R \square$
 $\mathbb{1} \sqsubseteq (R/\exists_R)(\exists_R/R)$
 $\frac{\exists_R}{\exists_R} = \mathbb{1}$

$\exists_R$ is **thick**
 $\exists_R$ is **straight**

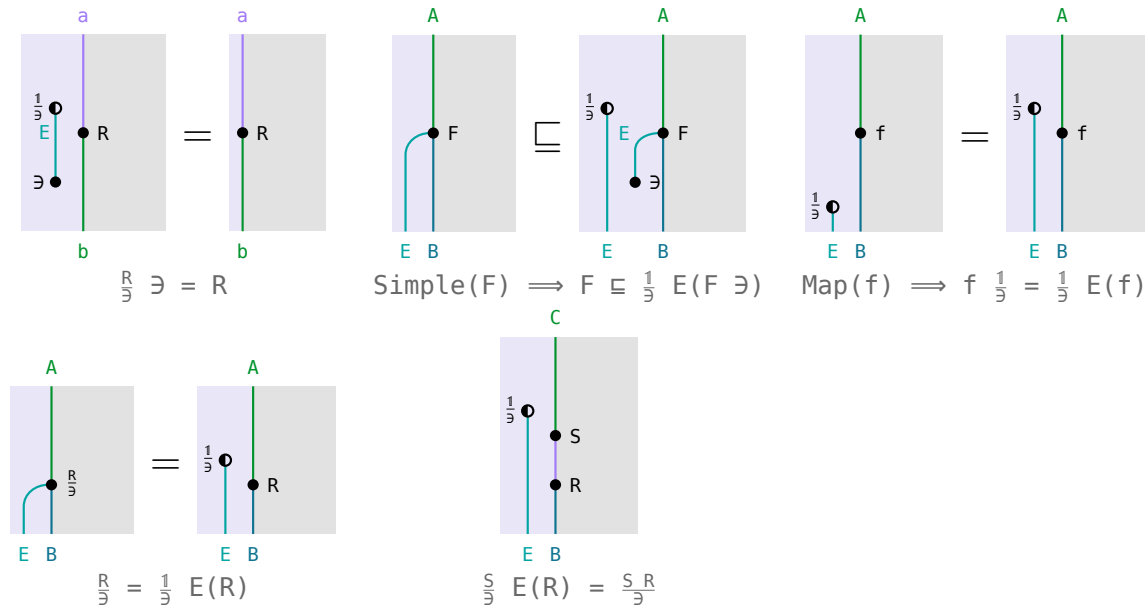
$R \square$ is R ’s target, an identity arrow. For $R : A \rightarrow B$ write $\exists : PB \rightarrow B$, dropping the subscript.

$\in \triangle \exists^\circ : A \rightarrow PA$

(10b)

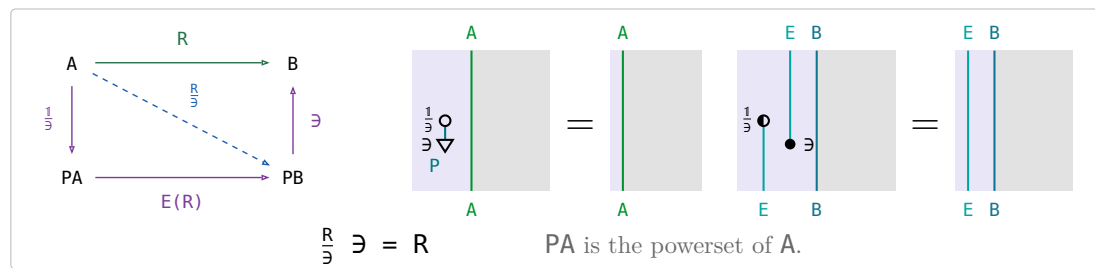
law	the reading
1 $\exists_R \square = R \square, \quad \exists_R = \exists_R \square$	One $\exists$ per object, not per arrow.
2 $\exists$ is thick	Comprehension: every x has a set of exactly the people x admires.
3 $\frac{R}{\exists} : A \rightarrow PB$, for $R : A \rightarrow B$	convert a relation to a function. $a \frac{R}{\exists} = \{b \mid a R b\}$
4 $\text{Map}(\frac{R}{\exists})$	
5 $\frac{R}{\exists} \exists = R$	reading the chosen set back through $\exists$ returns the relation.
6 $\text{Simple}(F) \implies F \sqsubseteq \frac{F \exists}{\exists}$	A partial choice of sets is inside the total one.
7 the singleton map is monic: $\frac{1}{\exists} (\frac{1}{\exists})^\circ \sqsubseteq \mathbb{1}$ two points with the same one-person set are the same point.	The one-person set.
8 $\frac{\exists}{\exists} = \mathbb{1}$	Make the set of a set, then read it back one level down.
9 fusion: $\text{Map}(f) \implies \frac{f R}{\exists} = f \frac{R}{\exists}$	Naturality of the unit, $\text{Map}(f) \implies f \frac{1}{\exists} = \frac{1}{\exists} E(f)$. renaming a point and then taking its one-person set is taking the set and renaming inside it.
10 $\text{Map}(f) \wedge \mathbb{1} \square = \exists \square \implies \frac{f}{\exists} = f \frac{1}{\exists}$	Rename first or take singletons first — the fusion row above at $R=1$.
11 $R \square = \exists \square \implies R /_s S = \frac{R}{\exists} (\frac{S}{\exists})^\circ$	x and y match when R sends x and S sends y to the same set.
12 $E(R) \triangleq \frac{\exists R}{\exists}$	$E(R) : PA \rightarrow PB$, image of a set of A
13 $\frac{R}{\exists} = \frac{1}{\exists} E(R)$	$\frac{1}{\exists} : x \mapsto \{x\}$
14 $\frac{S}{\exists} E(R) = \frac{S R}{\exists}$	absorption — the monad’s composition law, $\frac{S}{\exists} \diamond \frac{R}{\exists} = \frac{S R}{\exists}$, §Section 10.3
15 $\supseteq \triangleq \exists / \exists$	$xs \supseteq ys \iff \forall a. \quad ys \exists a \rightarrow xs \exists a$
16 $\subseteq = \supseteq^\circ$	$xs \subseteq ys \iff ys \supseteq xs$

(10c)



§10.1. $\mathbf{i} \vdash \mathbf{E}$ Power Algebra defined as adjunction

(10.1a)



$\mathbf{i}$ is the inclusion $\text{Map}(\mathcal{A}) \rightarrow \mathcal{A}$, doing nothing, so $\mathbf{E}$ is both the functor and the monad $\mathbf{i} \vdash \mathbf{E}$.

§10.2. $\mathbf{E} \setminus$ as a composite

$$\mathbf{E} \setminus \mathbf{Z} = \subseteq (\frac{\mathbf{Z}^\circ}{3})^\circ \quad (10.2a)$$

$\mathbf{x}\mathbf{s}$ is related to $\mathbf{c}$ by $\mathbf{E} \setminus \mathbf{Z}$ exactly when $\mathbf{x}\mathbf{s} \subseteq \Lambda(\mathbf{Z}^\circ)(\mathbf{c})$, the set of every $\mathbf{a}$ with $\mathbf{a} \mathbf{Z} \mathbf{c}$

(a)

(b)

(c)

(a)	$\mathbf{E} \setminus \mathbf{Z}$	$\mathbf{Z} = \mathbf{E} \Lambda(\mathbf{Z}^\circ)^\circ$ — (10b)
(b)	$\mathbf{E} \setminus \mathbf{E}(\frac{\mathbf{Z}^\circ}{3})^\circ$	$\mathbf{X} \setminus (\mathbf{Y} \mathbf{f}^\circ) = (\mathbf{X} \setminus \mathbf{Y}) \mathbf{f}^\circ$, $\mathbf{f}$ a map
(c)	$\subseteq (\frac{\mathbf{Z}^\circ}{3})^\circ$	

$$\mathbf{R} \setminus \mathbf{S} = \frac{\mathbf{R}^\circ}{3} \subseteq (\frac{\mathbf{S}^\circ}{3})^\circ \quad (10.2b)$$

$\mathbf{b}$ is related to $\mathbf{c}$ by $\mathbf{R} \setminus \mathbf{S}$ exactly when $\Lambda(\mathbf{R}^\circ)(\mathbf{b}) \subseteq \Lambda(\mathbf{S}^\circ)(\mathbf{c})$, the $\mathbf{R}$ -preimage of $\mathbf{b}$ inside the $\mathbf{S}$ -preimage of $\mathbf{c}$

(a)

(b)

(c)

(d)

(a)	$\mathbf{R} \setminus \mathbf{S}$	$\mathbf{R} = \mathbf{E} \Lambda(\mathbf{R}^\circ)^\circ$ — (10b)
(b)	$\mathbf{E}(\frac{\mathbf{R}^\circ}{3})^\circ \setminus \mathbf{S}$	$(\mathbf{X} \mathbf{Y}) \setminus \mathbf{S} = \mathbf{Y} \setminus (\mathbf{X} \setminus \mathbf{S})$
(c)	$(\frac{\mathbf{R}^\circ}{3})^\circ \setminus (\mathbf{E} \setminus \mathbf{S})$	$\mathbf{f}^\circ \setminus \mathbf{X} = \mathbf{f} \mathbf{X}$; $\mathbf{E} \setminus \mathbf{S} = \subseteq \Lambda(\mathbf{S}^\circ)^\circ$ — (10.2a)
(d)	$\frac{\mathbf{R}^\circ}{3} \subseteq (\frac{\mathbf{S}^\circ}{3})^\circ$	

§10.3. $\mathcal{A} \cong \mathbf{Kleisli}(E)$

(10.3a)

$E(R) \triangleq \frac{\exists}{\exists} R$, $\frac{1}{\exists} : A \rightarrow PA$, $\text{union} = E(\exists) : PPA \rightarrow PA$

$f \diamond g \triangleq f E(g) \text{ union}$, for $f : A \rightarrow PB$ and $g : B \rightarrow PC$

the monad is on $\mathbf{Map}(\mathcal{A})$, not on the allegory: E is a relator on all relations, but $\frac{1}{\exists}$, union and $f E(g) \text{ union}$ are maps, and the Kleisli construction happens where they live.

(10.3b)

$$\begin{array}{c}
 \boxed{\frac{S}{\exists} \diamond \frac{R}{\exists}} \xrightarrow{\text{definition of } \diamond, \text{ union} = E(\exists)} \boxed{\frac{S}{\exists} E(\frac{R}{\exists}) E(\exists)} \xrightarrow{E \text{ a functor, } E(X)E(Y) = E(XY)} \boxed{\frac{S}{\exists} E(\frac{R}{\exists} \exists)} \\
 \xrightarrow{(10b), \frac{R}{\exists} \exists = R} \boxed{\frac{S}{\exists} E(R)} \xrightarrow{(10b), \text{ absorption}} \boxed{\frac{SR}{\exists}}
 \end{array}$$

the first three steps only unfold $\diamond$ — E is a functor, so $E(\frac{R}{\exists}) \text{ union} = E(\frac{R}{\exists} \exists) = E(R)$ and the union is gone.

Absorption, the row of (10b), is the whole law, and it is the functoriality of $\frac{1}{\exists}$.

1 goes to $\frac{1}{\exists}$, the Kleisli identity, by definition — so $\frac{1}{\exists}$ is an isomorphism of categories $\mathcal{A} \cong \mathbf{Kleisli}(E)$, and §Section 10.1's $i \dashv E$ is the Kleisli adjunction. i is the identity on objects, so every object of the allegory is free.